

When Historical Predictions Become Stale: Robust Expert Combination under Temporal Behavioral Shift

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Independent Research Study in Machine Learning and Online Algorithms

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github.com/ngclara07/robust_prediction_learning

September 22, 2026

Abstract

Machine-learning systems deployed on sequential behavioral data frequently rely on predictors trained from historical observations. Such predictors may become unreliable when user behavior changes over time. This study investigates when historical predictions should be trusted and whether behavioral distribution shift can provide a useful signal for robust prediction combination.

We study this problem using the Last.fm-1K music-listening dataset. After preprocessing, the experimental corpus contains approximately 18.5 million timestamped interactions from 966 users and 41,045 artists. A Bayesian Personalized Ranking (BPR) model is trained on historical implicit feedback and evaluated under chronological splits. Behavioral shift is quantified using Jensen–Shannon divergence between historical artist distributions and equal-interaction future windows.

Across 6,642 validation windows from 595 sufficiently active users, behavioral shift is negatively associated with frozen historical-model quality: the pooled Spearman correlation between shift and EventMass@20 is -0.359 , the median within-user correlation is -0.252 , and 70.1% of users with valid correlations exhibit a negative relationship. This qualitative result remains stable for window sizes of 125, 250, and 500 interactions.

We subsequently compare five sequential strategies: historical prediction only, online adaptation only, a static mixture, Hedge, and a shift-aware adaptive trust mechanism. On the held-out test stream, online adaptation achieves the highest expected EventMass@20 (0.358), followed by Hedge (0.319). The proposed shift-aware mechanism reaches 0.240, outperforming exclusive reliance on the historical predictor (0.063) but not generic loss-adaptive aggregation.

The results indicate that behavioral shift is informative about historical-predictor degradation, while also showing that an explicit shift signal alone is insufficient to outperform rapid online adaptation in the studied setting. The negative result identifies an important distinction between detecting distribution shift and optimally responding to it.

1 Introduction

Machine-learning models are commonly trained on historical data and then deployed under the implicit assumption that future observations remain sufficiently similar to their training distribution. In sequential applications, this assumption is often violated. User preferences evolve, new items appear, and environmental conditions change. Consequently, a predictor that was previously accurate can become stale.

This problem is related to concept drift and non-stationary learning, where the statistical properties relevant to prediction change over time [Gama et al., 2014]. It is also related to the

emerging literature on *algorithms with predictions*, which studies how classical algorithms can benefit from machine-learned advice while remaining robust when that advice is inaccurate [Lykouris and Vassilvitskii, 2018, Mitzenmacher and Vassilvitskii, 2022].

Recommender systems provide a useful real-world testbed for this problem. Historical listening behavior can be used to estimate personalized preferences, but musical interests can change substantially over time. Furthermore, recommendation data are typically implicit, sparse, long-tailed, and temporally structured [Adomavicius and Tuzhilin, 2005, Celma, 2010b].

This study asks the following central research question:

Can behavioral distribution shift be used to determine when a sequential learning algorithm should reduce its trust in a historical machine-learning predictor?

We first establish whether measurable behavioral shift is associated with deterioration in a historical recommender. We then formulate the problem as online combination of two experts:

- (i) a frozen historical BPR predictor; and
- (ii) a lightweight online predictor that adapts to recent interactions.

Several combination strategies are compared, including a fixed convex mixture, the Hedge algorithm, and a shift-aware adaptive trust mechanism.

The principal contributions of this independent study are:

1. a reproducible chronological evaluation pipeline for studying historical-predictor degradation on approximately 18.5 million Last.fm listening events;
2. an empirical analysis showing that larger Jensen–Shannon divergence from historical listening distributions is consistently associated with lower historical-predictor quality;
3. a robustness analysis demonstrating that this association remains qualitatively stable across multiple temporal window sizes;
4. an online expert-combination formulation for comparing prediction-only, adaptation-only, static, loss-adaptive, and distribution-shift-aware strategies; and
5. a negative but informative result: the proposed shift-aware trust rule improves substantially over exclusive reliance on stale historical predictions, but does not outperform either pure online adaptation or Hedge under the present formulation.

This last result is particularly important. Detecting that a distribution has changed is not equivalent to knowing how much historical information should be retained. The experiments therefore suggest that reliable prediction augmentation requires both a meaningful shift signal and an effective adaptation mechanism.

2 Related Work

2.1 Implicit-Feedback Recommendation

Recommender systems estimate user preferences from observed interaction data [Adomavicius and Tuzhilin, 2005]. Music recommendation is particularly challenging because listening behavior is highly long-tailed and repeated consumption is common [Celma, 2010b].

Bayesian Personalized Ranking (BPR) was introduced for learning personalized rankings from implicit feedback [Rendle et al., 2009]. Given a user u , an observed item i , and an unobserved item j , BPR learns parameters that increase the pairwise preference score

$$\hat{x}_{uij} = \hat{x}_{ui} - \hat{x}_{uj}. \quad (1)$$

For matrix factorization, user and item representations are optimized using a pairwise logistic objective.

In this study, BPR is not intended as a state-of-the-art recommender. Instead, it serves as a learned historical predictor whose reliability can be studied as behavior changes.

2.2 Temporal Distribution Shift

Concept drift describes settings in which relationships relevant to prediction evolve over time [Gama et al., 2014]. Such changes can invalidate models trained under earlier distributions and motivate continuous adaptation.

We quantify behavioral change using Jensen–Shannon divergence [Lin, 1991]. For two discrete distributions P and Q , let

$$M = \frac{1}{2}(P + Q). \quad (2)$$

The Jensen–Shannon divergence is

$$\text{JS}(P, Q) = \frac{1}{2}D_{\text{KL}}(P\|M) + \frac{1}{2}D_{\text{KL}}(Q\|M). \quad (3)$$

Using logarithms base two yields a bounded quantity in $[0, 1]$.

2.3 Online Learning and Algorithms with Predictions

Online learning studies sequential decision-making when outcomes are revealed after predictions are made. The Hedge algorithm uses multiplicative weight updates to allocate increasing weight to experts with lower observed loss [Freund and Schapire, 1997].

For expert i with loss $\ell_{i,t}$, Hedge performs an update of the form

$$w_{i,t+1} \propto w_{i,t} \exp(-\eta \ell_{i,t}), \quad (4)$$

where $\eta > 0$ is a learning-rate parameter.

More recent work on algorithms with predictions investigates how machine-learned advice can improve algorithmic decisions without sacrificing robustness when predictions fail [Lykouris and Vassilvitskii, 2018, Mitzenmacher and Vassilvitskii, 2022]. The present study adopts this perspective: historical BPR scores are treated as learned advice rather than as an unquestioned source of truth.

3 Dataset and Preprocessing

3.1 Last.fm-1K

Experiments use the Last.fm-1K music recommendation dataset collected by Celma [2010a] from the Last.fm service [Last.fm]. The original data contain timestamped listening events with user, artist, and track information.

The study operates at the artist level. Artist names are normalized and used as operational item identifiers because MusicBrainz identifiers are missing for a substantial portion of the original records.

Users with fewer than 100 retained interactions and artists with fewer than 20 retained interactions are filtered. After preprocessing, the dataset contains the statistics shown in Table 1.

Table 1: Preprocessed dataset statistics.

Statistic	Value
Raw listening events	19,098,853
Processed listening events	18,521,447
Users	966
Artists	41,045
Earliest timestamp	14 February 2005
Latest timestamp	29 September 2013
Training interactions	12,964,572
Validation interactions	1,852,070
Test interactions	3,704,805

A per-user chronological split is used:

70% training, 10% validation, 20% test.

No random interaction shuffling is performed. For each user, all training events precede validation events, which in turn precede test events.

The preprocessing pipeline automatically verifies this temporal ordering to prevent future-to-past leakage.

3.2 Exploratory Characteristics

The artist-frequency distribution is strongly long-tailed: a small number of artists account for very large interaction volumes while many artists have relatively few interactions. User activity is also highly heterogeneous.

The global interaction timeline contains an abrupt reduction in observed events after approximately 2009. This pattern is treated as a dataset-coverage artifact rather than direct evidence of behavioral change. Consequently, our primary drift analysis does not use fixed calendar months. Instead, equal-interaction windows are constructed separately for each user.

4 Historical Recommendation Model

4.1 BPR Training

Repeated listening events are collapsed into unique user–artist positive pairs when fitting BPR. The historical model therefore learns implicit preference from whether a user has interacted with an artist, rather than allowing repeated plays alone to dominate optimization.

The selected BPR configuration is:

Table 2: BPR hyperparameters selected using validation data.

Parameter	Value
Latent factors	64
Learning rate	0.05
Regularization	0.005
Epochs	5
Batch size	8,192
Maximum sampled pairs per epoch	500,000

4.2 Chronological Baseline

Table 3 compares BPR against a global popularity baseline on the chronological test split.

Table 3: Chronological recommendation baseline performance.

Model	Precision@20	Recall@20	NDCG@20	MRR@20
Popularity	0.0886	0.0242	0.0976	0.2462
BPR	0.0894	0.0232	0.0937	0.2131

BPR is therefore useful as a personalized learned predictor but is not uniformly superior to popularity on the later test period. This observation motivates studying how predictive usefulness changes over time rather than assuming that historical personalization remains reliable indefinitely.

5 Temporal Behavioral Shift

5.1 Equal-Interaction Windows

For each user u , the validation interaction stream is divided chronologically into windows containing a fixed number of listening events. The primary analysis uses 250 interactions per window and requires at least three complete windows per user.

Let

$$P_u^H \tag{5}$$

denote the normalized historical artist distribution for user u , and let

$$P_{u,t} \tag{6}$$

denote the artist distribution in future window t .

Behavioral shift is defined as

$$s_{u,t} = \text{JS}(P_u^H, P_{u,t}). \tag{7}$$

5.2 EventMass@K

Repeated listening is meaningful in music consumption. We therefore supplement binary ranking metrics with EventMass@K.

For recommendation set $R_{u,t}^K$ and event counts $n_{u,t}(i)$,

$$\text{EventMass@K}_{u,t} = \frac{\sum_{i \in R_{u,t}^K} n_{u,t}(i)}{\sum_j n_{u,t}(j)}. \quad (8)$$

This measures the fraction of actual listening activity covered by the recommended top- K artists.

5.3 Shift and Predictor Quality

At the primary window size of 250 interactions, 595 users provide at least three complete validation windows, producing 6,642 windows.

The mean Jensen–Shannon divergence from historical behavior is 0.642, with median 0.663. The mean fraction of events involving artists globally unavailable to the historical model is only 1.03%, while approximately 14.1% of future events involve artists not previously observed for the corresponding user.

The pooled Spearman correlation between behavioral shift and historical BPR EventMass@20 is

$$\rho = -0.359. \quad (9)$$

For BPR NDCG@20, the pooled Spearman correlation is

$$\rho = -0.484. \quad (10)$$

Because windows belonging to the same user are not statistically independent, pooled significance tests are not treated as the main inferential evidence. Instead, we compute a separate Spearman correlation within each user. Among 562 users with valid within-user correlations, the median is

$$\text{median}(\rho_u) = -0.252, \quad (11)$$

and 70.1% have $\rho_u < 0$.

Figure 1 illustrates the relationship.

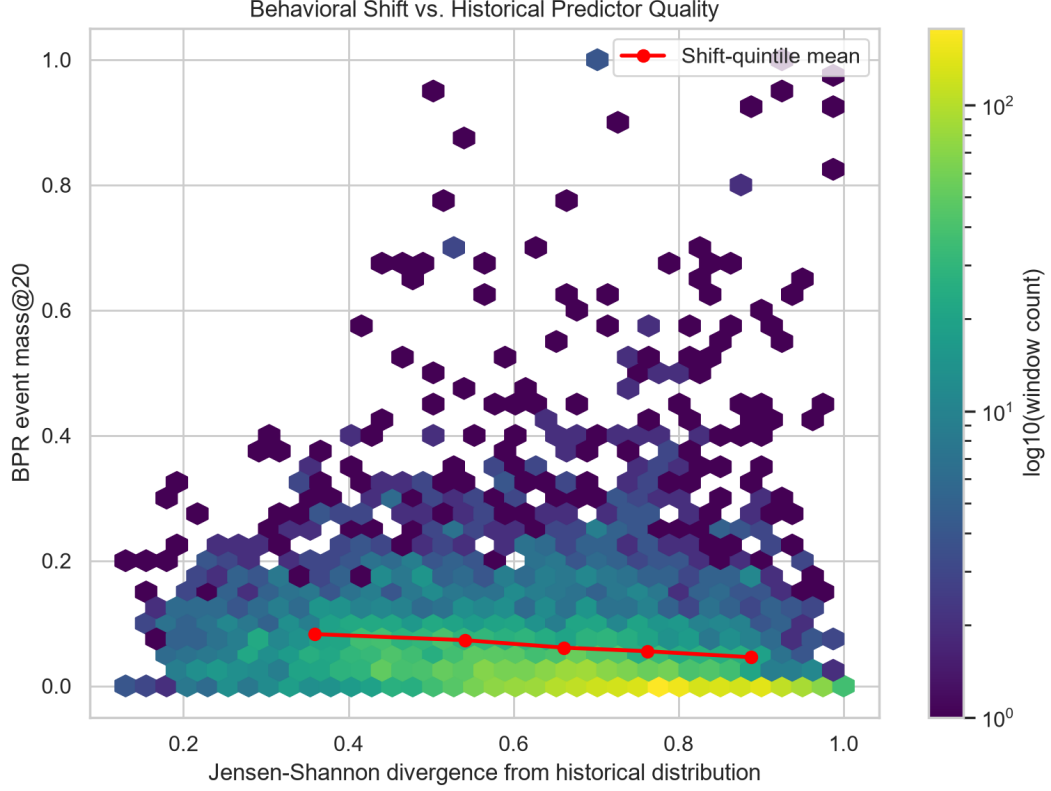


Figure 1: Behavioral shift versus frozen historical BPR quality. Hexagonal density cells summarize validation windows; the red line reports mean EventMass@20 across shift quintiles. Larger divergence is associated with lower historical-predictor coverage.

The distribution of within-user correlations is shown in Figure 2.

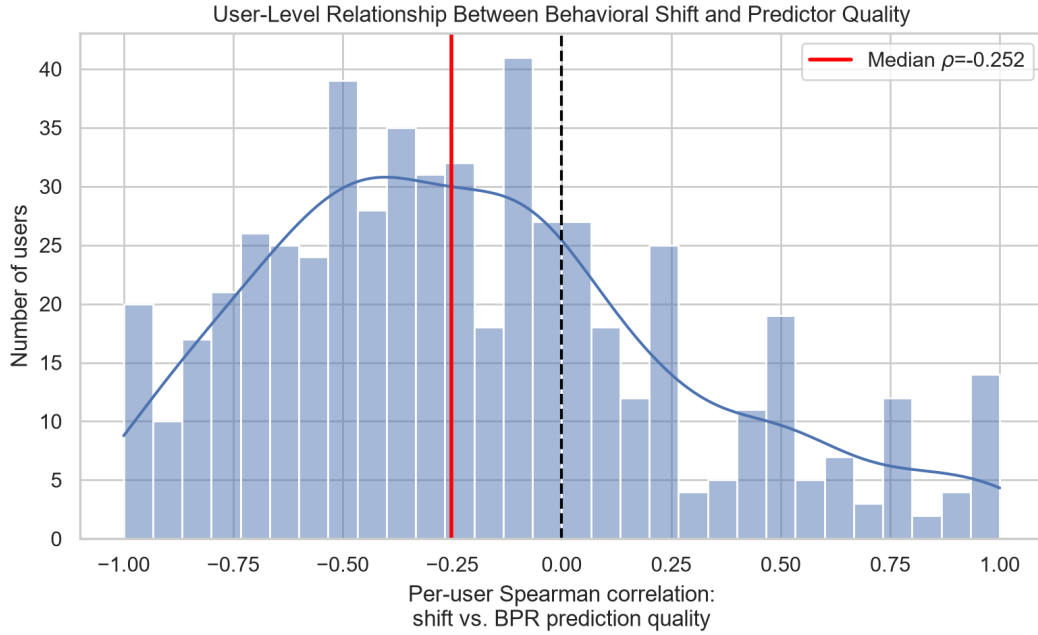


Figure 2: Distribution of per-user Spearman correlations between behavioral shift and historical BPR prediction quality. The median correlation is negative, although substantial user-level heterogeneity remains.

5.4 Window-Size Sensitivity

To determine whether the observed relationship depends strongly on the choice of 250 interactions per window, the analysis is repeated with window sizes of 125 and 500.

Table 4: Sensitivity of the shift–quality relationship to temporal window size.

Window	Users	Windows	Overall ρ	Median user ρ
125	736	14,153	-0.383	-0.284
250	595	6,642	-0.359	-0.252
500	405	2,848	-0.334	-0.300

The fraction of users with negative within-user correlation is 74.0%, 70.1%, and 66.1% for window sizes 125, 250, and 500 respectively. Thus, while quantitative values vary, the qualitative relationship is stable across all three temporal resolutions.

6 Robust Prediction Combination

6.1 Sequential Experts

The robust-learning experiment treats prediction as a combination of two experts.

The *historical expert* is a frozen BPR model trained before the evaluation stream.

The *online expert* maintains exponentially decayed recent interaction counts:

$$c_{u,t}(i) = \lambda c_{u,t-1}(i) + n_{u,t}(i), \quad (12)$$

where $\lambda = 0.8$ and $n_{u,t}(i)$ denotes interactions with artist i in the current window. A global-popularity prior is used before sufficient online observations have accumulated.

Critically, predictions for window t are produced before that window’s observations are incorporated into the online model.

6.2 Loss

For expert e ,

$$\ell_{e,t} = 1 - \text{EventMass@20}_{e,t}. \quad (13)$$

Thus,

$$0 \leq \ell_{e,t} \leq 1. \quad (14)$$

For mixture weight α_t on the historical expert, the expected mixture loss is

$$\ell_t = \alpha_t \ell_{H,t} + (1 - \alpha_t) \ell_{O,t}. \quad (15)$$

This quantity corresponds to the expected loss of probabilistically selecting the historical expert with probability α_t and the online expert otherwise.

6.3 Methods

Five strategies are compared.

Prediction Only.

$$\alpha_t = 1. \quad (16)$$

Online Only.

$$\alpha_t = 0. \quad (17)$$

Static Mixture.

$$\alpha_t = \alpha, \quad (18)$$

where α is selected using validation data.

Hedge. The historical and online predictors are treated as experts and updated using exponentially weighted losses [Freund and Schapire, 1997].

Shift-Aware Adaptive Trust. The proposed candidate mechanism combines behavioral shift and recent relative expert performance. Let

$$d_t = (1 - \beta)d_{t-1} + \beta(\ell_{H,t} - \ell_{O,t}) \quad (19)$$

be an exponentially smoothed historical disadvantage.

Before the current window is observed, historical trust is

$$\alpha_t = \sigma(\text{logit}(\alpha_0) - \gamma_s s_t - \gamma_d d_{t-1}), \quad (20)$$

where σ is the logistic function, s_t is the current behavioral-shift signal computed from previously observed data, α_0 is baseline trust, and γ_s, γ_d control sensitivity to shift and recent relative loss.

All adaptive parameters are selected on the validation stream before final test evaluation.

7 Robust Learning Results

The final historical models are fitted using training and validation data. Robust-learning parameters are frozen before the held-out test stream is evaluated.

The selected validation parameters include a static historical weight of 0.25, Hedge learning rate $\eta = 2.0$, and shift-aware parameters

$$\alpha_0 = 0.75, \quad \gamma_s = 2.0, \quad \gamma_d = 2.0, \quad \beta = 0.25.$$

A total of 736 users satisfy the requirement of at least three complete 250-interaction test windows.

Table 5 reports final performance.

Table 5: Held-out sequential robust-learning results. Lower loss and regret are better; higher expected EventMass@20 is better.

Method	Mean loss	Expected EventMass@20	Mean regret
Online Only	0.6421	0.3579	0.0004
Hedge	0.6815	0.3185	0.4179
Static Mixture	0.7158	0.2842	1.5119
Shift-Aware Trust	0.7599	0.2401	2.2557
Prediction Only	0.9369	0.0631	6.0464

Figure 3 visualizes expected EventMass@20.

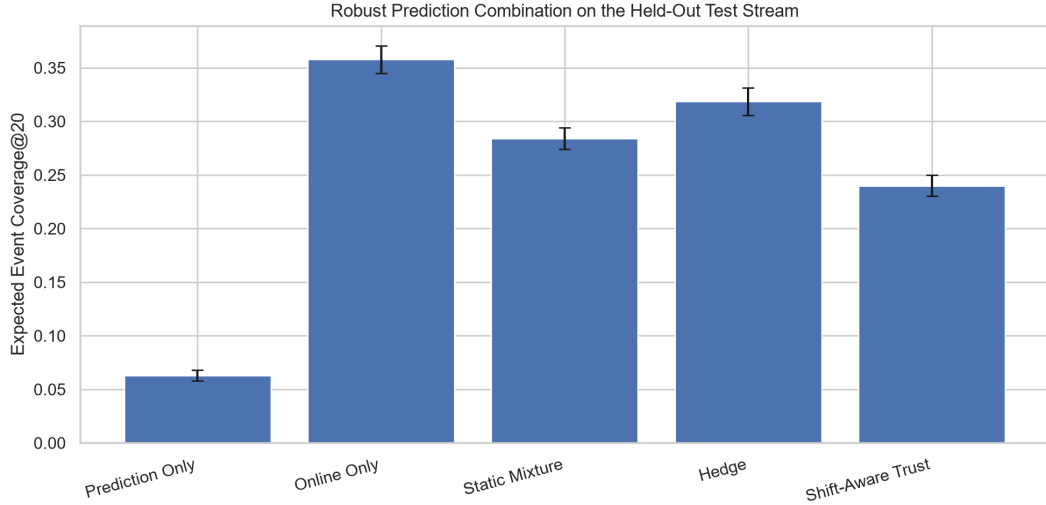


Figure 3: Final robust-learning comparison on the held-out sequential test stream. Error bars show 95% normal-approximation intervals over user-level mean losses.

7.1 Historical Prediction Becomes Weak

Prediction Only performs substantially worse than every strategy that incorporates online evidence. Its expected EventMass@20 is only 0.0631.

This result is consistent with the shift analysis: historical personalization can become severely stale when future behavior differs from the historical distribution.

7.2 Online Adaptation Dominates

Online Only achieves the strongest average performance with expected EventMass@20 of 0.3579 and near-zero mean regret relative to the better fixed expert.

This implies that, under the current experimental design, the recent interaction stream contains substantially more useful information than the frozen historical BPR predictor.

Hedge provides the strongest combined-expert strategy, obtaining EventMass@20 of 0.3185. Its multiplicative update rapidly decreases the weight assigned to BPR once the online expert demonstrates lower loss.

7.3 Why Shift-Aware Trust Underperforms

The proposed shift-aware method improves substantially over Prediction Only but does not outperform Static Mixture, Hedge, or Online Only.

Figure 4 provides a mechanistic explanation.

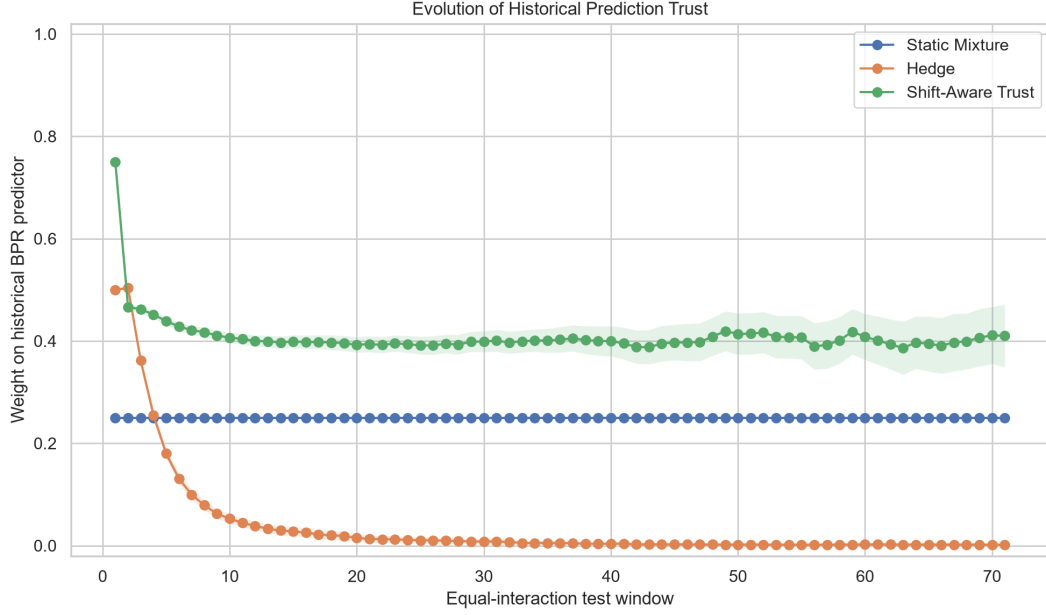


Figure 4: Mean weight placed on the frozen historical BPR predictor across sequential test windows. Hedge rapidly reduces historical trust toward zero, whereas the selected shift-aware rule stabilizes around a considerably larger historical weight.

The selected shift-aware method retains approximately 0.4 historical weight over much of the stream. Hedge, by contrast, rapidly learns to place almost no weight on BPR.

The negative result therefore does not imply that the behavioral shift signal contains no information. Milestone 3 demonstrates that shift is associated with predictor degradation. Instead, the result shows that the current mapping

$$s_t \longrightarrow \alpha_t$$

does not withdraw historical trust aggressively enough relative to the strength of the online expert.

Figure 5 further shows performance as behavioral shift increases.

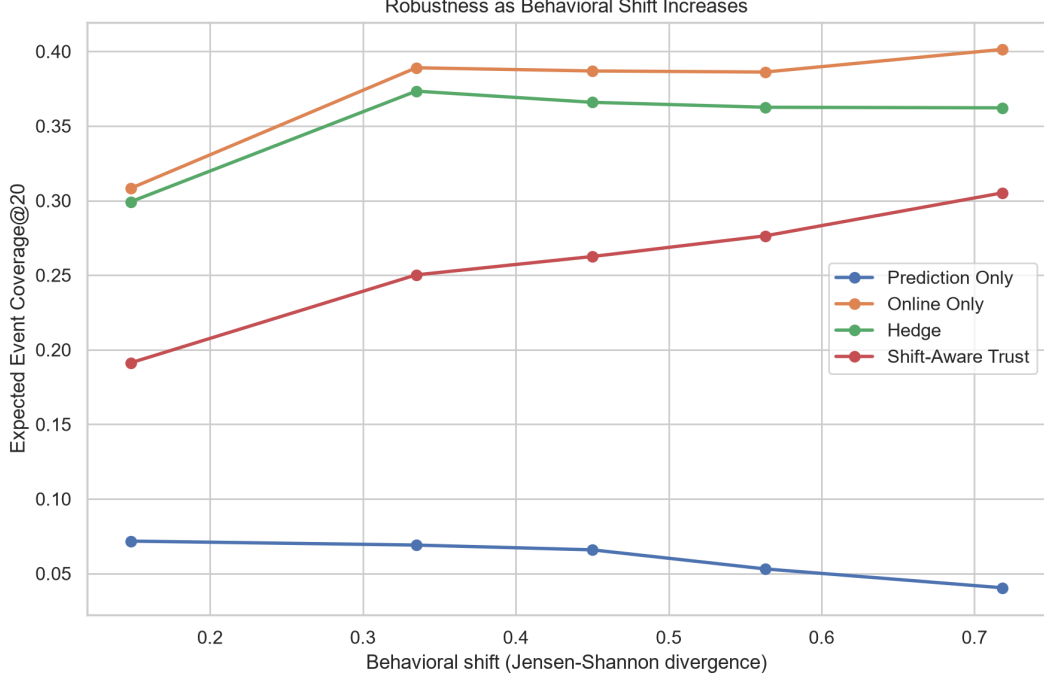


Figure 5: Expected EventMass@20 across behavioral-shift quintiles. The historical predictor deteriorates as shift increases. Online adaptation remains strongest across the observed shift range, while Hedge provides the strongest mixed-expert strategy.

8 Discussion

The experiments support three principal conclusions.

First, temporal behavioral shift is measurable and associated with historical-model degradation. This relationship appears both in pooled analysis and, importantly, within individual users. Its qualitative stability across several window sizes suggests that it is not merely an artifact of one arbitrary temporal resolution.

Second, detecting shift and responding optimally to shift are different problems. Jensen–Shannon divergence contains information about historical-predictor reliability, yet the explicit shift-aware gating rule does not outperform loss-adaptive Hedge.

Third, the strong Online Only result demonstrates that historical predictions need not always be beneficial. The algorithms-with- predictions perspective often seeks a balance between exploiting accurate advice and remaining robust when advice fails. In this particular dataset and evaluation stream, the learned advice becomes weak enough that rapid adaptation is generally preferable.

This negative result is useful for future algorithm design. A more competitive shift-aware method may require a mechanism capable of distinguishing between:

- transient behavioral variation;
- persistent distribution shift;
- regions in which historical predictions remain useful; and
- cases where the online predictor has accumulated sufficient evidence to supersede historical information.

A single global logistic mapping from divergence to trust may be too restrictive.

9 Limitations

Several limitations constrain the conclusions.

Single dataset. The study evaluates one historical music-consumption dataset. Results should not be interpreted as universal properties of modern recommender systems.

Historical data collection. The Last.fm observations are not representative of current music consumption. The experiments study algorithmic behavior under temporal shift rather than contemporary popularity.

Active-user selection. Equal-interaction windows require sufficiently active users. At the primary setting, only 595 validation users and 736 test users satisfy the minimum-window criterion. Conclusions therefore emphasize active listeners.

Artist-level identifiers. Normalized artist names are used as operational item identifiers. Name collisions or spelling inconsistencies may introduce noise.

Observational shift analysis. The negative association between Jensen–Shannon divergence and historical-predictor quality does not establish a causal relationship.

Simple online expert. The online model uses exponentially decayed interaction counts rather than a sophisticated sequential neural recommender. This simplicity is intentional for isolating the expert-combination problem, but limits the scope of conclusions about recommendation architectures.

Mixture interpretation. Combined strategies are evaluated by expected convex mixture loss. This corresponds to randomized expert selection rather than explicitly constructing a new deterministic top- K ranking from heterogeneous score spaces.

Initial shift-aware formulation. The proposed trust mechanism constitutes an initial candidate rather than a theoretically optimal algorithm. Its underperformance should be interpreted as evidence about the formulation, not as evidence that distribution-shift signals cannot be useful.

10 Conclusion

This study investigated robust use of historical machine-learning predictions under temporal behavioral shift.

Using approximately 18.5 million processed Last.fm interactions, we found a consistent negative association between divergence from historical listening behavior and the quality of a frozen BPR

predictor. The result persisted across several equal-interaction window sizes and was observable within a majority of individual users.

We then formulated sequential recommendation as an online expert-combination problem. Exclusive reliance on historical prediction performed poorly, while online adaptation achieved the strongest held-out performance. Hedge successfully learned to discount the stale historical predictor and was the strongest mixed-expert strategy.

A shift-aware adaptive trust rule improved substantially over Prediction Only but did not outperform Hedge or Online Only. Inspection of its learned trust trajectory showed that it retained excessive historical weight after the online expert had become substantially more reliable.

The central conclusion is therefore twofold: behavioral distribution shift provides useful evidence about predictor reliability, but effective robustness requires more than detecting that shift. Future work should investigate theoretically grounded mechanisms that convert local distribution-shift evidence into prediction trust while retaining the worst-case robustness properties of online expert algorithms.

Reproducibility Statement

The complete preprocessing pipeline, recommendation baselines, distribution-shift analysis, robust-learning algorithms, unit tests, figures, and experiment scripts are maintained in the accompanying research repository.

All major experiments use fixed random seeds. Hyperparameters for BPR and robust expert-combination methods are selected using validation data. Final robust-learning comparisons use the held-out chronological test stream.

The Last.fm source data are not redistributed in the repository and must be obtained separately under the dataset’s applicable non-commercial-use conditions.

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